

Commission calculator final check

2 October 2026. Unit tests on the current solution, then a live pass against the rebuilt Docker stack on ports 8081, 8082, and 8083. Edge-case prefix chk222528. Load prefix load223525, with two workers of each kind.

8/10 Scaling, overall	29/29 Unit tests passed	43/45 Live checks matched the rules	1 Real edge-case failure
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Scaling scores out of 10

Before is the single-Postgres review. After is this codebase, from the edge-case pass and the two-worker load run at 22:35 local.

Axis	Before	After
Replica safety	8	9
Throughput	4	7
Compose and Postgres	3	7
Overall	6	8

Replica safety, 9

HTTP processes do not run workers. Outbox, pending events, and payout batches use FOR UPDATE SKIP LOCKED. A payout lease expires after 90 seconds and keeps the same payout id. Commission ids on wallet lines are unique.

Throughput, 7

Extra workers can take different batches, up to 20 items, then sleep. Each service still has one Postgres primary, so more containers do not add write capacity past that primary.

Compose and Postgres, 7

Three databases, pool 20, max_connections 100. About four processes per database approach the connection ceiling. API replicas still need an external load balancer because each published port is one host port.

What still limits the score. Referrer writes stay Serializable, so a hot branch still queues in Postgres. The load run shortened worker intervals so the race finished in seconds. The shipped payout interval is still 60 seconds.

Unit tests

dotnet test CommissionCalculator.sln, 2 Oct 2026, 22:20 local. Test time under 50 ms.

Project	Passed	Failed	What it covers
Users.UnitTests	7	0	Self, cycle, depth 10, reparent, case
Accrual.UnitTests	22	0	Formulas, intake, claim idempotency

Project	Passed	Failed	What it covers
Wallet	0	0	No wallet unit tests in the solution

Live checks

Images were rebuilt from the current source before the run. Health ready returned 200 on all three services. The existing volume still had schema Fibonacci from an earlier session. The migration seed for an empty database is Linear, and this run set the schema back to Linear before the money cases. It is Linear again now.

Area	Result	Observed
Ids and replay of a user	Pass	128 chars 201, 129 and empty 400, same id 200, different case 201
Tree	Pass	Self 400, missing user and referrer 404, cycle 400, up and down match the chain
Depth	Pass	10 ancestors accepted, 11th 400, reparent past 10 400, reparent that stays at 10 204
Linear 1000	Pass	mid level 1 gets 10, root level 2 gets 20, schema Linear
Replay and conflict	Pass	Same body 200, other profit 409, other user 409
Non-positive profit	Pass	0 and -15 stored, zero commissions, both listed
Unknown user	Pass	POST 404 and the event is not stored
Rounding	Pass	Profit 0.4 omitted. Profit 0.5 pays 0.01, midpoint away from zero
Fibonacci snapshot	Pass	New event mid 10 and root 10, Fibonacci. Older event stays Linear
Bad schema	Pass	Unknown enum 400. Schema restored to Linear
Referrer delete	Pass	204 and up is empty. Missing user 404
Wallet read	Pass	First balance 0. Invalid id 400. Correlation id echoed
Payout	Pass	mid 20.01, root 30.03, paid true, second read unchanged

Worker load, prefix load223525

Schema was set to Linear. Wallet workers were stopped, then 48 events and 10 copies of one event were posted together. Two accrual workers delivered the outbox. Two wallet workers were started only after all 49 inbox rows were Received. Workers were returned to one replica and the default intervals afterward.

Check	Result	Observed
Parallel accepts	Pass	48 of 48 returned 201 in 361 ms
Same event, 10 at once	Pass	1 created, 9 returned 200, 1 event row
Outbox	Pass	49 of 49 Delivered, none left pending
Inbox	Pass	49 of 49 Completed, no Claiming or Received left
Paid once	Pass	49 commissions, 49 paid, 0 duplicate commission ids
Money	Pass	Lines sum to 490.00 and the root balance is 490.00
Two payout replicas	Pass	Worker 1 booked 200 and 90. Worker 2 booked 200. Same second, 3 payout ids

Check	Result	Observed
Leftover leases	Pass	0 payout batches remained

From the first post to every line paid: 6.0 seconds, with the payout interval set to 2 seconds for this run only.

Edge case that failed

Profit with more than 4 decimal places is not idempotent. POST profit 1.23456 returned 201 and stored 1.2346, because profit is numeric(19,4). The same JSON body then returned 409, because the stored profit no longer matches the request. Commissions were still correct for the stored value: level 1 gets 0.01, level 2 gets 0.02, and both lines were paid.

Why the payout totals are 20.01 and 30.03

Event	mid	root
Linear profit 1000	10	20
Fibonacci profit 1000	10	10
Linear profit 0.5 on mid	0	0.01
Stored profit 1.2346	0.01	0.02
Booked total	20.01	30.03

History lines: mid 3, root 4. A second balance read 8 seconds later was unchanged, so the worker did not book the same commissions again.